

Complete NestJS Learning Roadmap (From Zero to Project)

This document provides a complete learning roadmap for NestJS. It assumes you have no prior knowledge of NestJS. The final goal is to be able to build a simple and clean REST API project using NestJS best practices.

Stage 0: Prerequisites

- Basic JavaScript (variables, functions, async/await)
- Basic TypeScript (type, interface, class)
- Basic understanding of HTTP, REST and APIs

Stage 1: Introduction to NestJS

- What is NestJS and why it exists
- NestJS architecture overview
- Difference between NestJS and Express

Stage 2: Installation and First Project

- Install Node.js
- Install NestJS CLI
- Create a project using nest new
- Understand project folder structure

Stage 3: Core Concepts

- What is a Module
- What is a Controller
- What is a Service
- Dependency Injection (DI)

Stage 4: Routing and Request Handling

- GET, POST, PUT, DELETE
- Route parameters
- Query parameters
- Request body

Stage 5: DTO and Validation

- What is DTO
- class-validator
- class-transformer
- ValidationPipe

Stage 6: Working with Database

- Connect NestJS to MySQL or PostgreSQL
- Using TypeORM or Prisma
- Entity
- Repository pattern

Stage 7: Error Handling

- HttpException
- Exception Filters
- Centralized error handling

Stage 8: Authentication and Authorization

- Authentication vs Authorization
- JWT
- Guards
- Protected routes

Stage 9: Practice Project

- Build a simple REST API
- User registration and login
- Basic CRUD operations
- Database integration

Stage 10: Preparing for Real Projects

- Clean project structure
- Environment variables
- Dockerizing NestJS application
- Best practices and optimization

Conclusion By following this roadmap, you will be able to build a clean and structured NestJS project with authentication, database integration, and scalable architecture suitable for real-world applications.